Jeff Brozena

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Education

2021– **Ph.D., Informatics**
NSF Graduate Research Fellow
The Pennsylvania State University • Advised by Dr. Saeed Abdullah
Research Interests: *Human-Computer Interaction, Mental Health Informatics, Digital Health,* and *Financial Technologies*

2019 **B.Sc., Information Sciences and Technology**
The Pennsylvania State University
Minor: Science, Technology, and Society Studies (STS)

Research Experience

2021– **Supportive Financial Technologies for Bipolar Disorder**
The Wellbeing and Health Innovation Lab, The Pennsylvania State University

Survey Experiments
- Factorial vignette studies assessing privacy preferences surrounding third-party financial data sharing and digital health interventions (adults with bipolar disorder, N=980)
- MaxDiff scaling of digital financial intervention feature preferences (adults with bipolar disorder, N=150)

Trust-based Clinical Population Recruitment and Data Collection (Adults with bipolar disorder, N=50)
- *Two continuous years of retrospective financial transactions* per participant
- Monthly anchored retrospective mood logs
- Semi-structured qualitative interviews

Multilevel Modeling and Time Series Analysis
- Multilevel analysis of financial data sharing privacy preferences in bipolar disorder
- Multilevel analysis of financial behaviors in relation to mood severity
- Vector autoregression of eight years of uninterrupted daily wearable-derived HRV anomalies and self-reported mood logs, no missing data

Qualitative Analysis
- Grounded theory analysis of semi-structured interviews

Professional Experience

2011–2017 **Director of Operations**
Smiles 4 Keeps Pediatric Dentistry

Led an end-to-end digital transformation at one of the country's largest privately-held multi-site Medicaid pediatric dental groups **while it continued to serve over 40,000 children annually**. Digitized all clinical and financial systems, trained 100+ staff on the redesigned environment, and rendered the business legible to third parties during due diligence phases. My work afforded Smiles 4 Keeps an accurate third-party valuation and **resulted in the sale of the business**. Invited to present to 5,000 pediatric dentists at the American Academy of Pediatric Dentistry conference on the subjects of digitization and compliant record-keeping.

Skills

Research Methods: Experimental Research, Survey Experiments (Factorial Vignettes, MaxDiff), Mixed-Methods Research, Human Subjects Research, Clinical Population Recruitment, Longitudinal Studies, Anchored Retrospective Recall, Wearable Sensing, Interviews

Data Analyses Techniques: Multilevel Modeling (*lme4*), Applied Time-Series Modeling (*mlVAR*), Marginal Effects Estimation (*marginaleffects*), Grounded Theory for Qualitative Data

Programming Languages, Tools, and Frameworks: R, Python, Latex, Django, PostgreSQL, RESTful APIs, Figma

Awards and Scholarships

2025 Sawtooth Software Academic Grant

2023 NSF Graduate Research Fellowship

2011 Rock Ethics Award, Pennsylvania State University

Publications

Preprints

2025 **Jeff Brozena**, Johnna Blair, Dahlia Mukherjee, Erika F. H. Saunders, Thomas Richardson, Mark Matthews, Saeed Abdullah. Assessing Acceptance and Privacy Preferences of Third-party Financial Data Sharing in Bipolar Disorder. In submission. https://doi.org/10.48550/arXiv.2610.04042.

2023 **Jeff Brozena**. Vector Autoregression (VAR) of Longitudinal Sleep and Self-report Mood Data. https://doi.org/10.48550/arXiv.2510.02511.

Conference Papers

2024 **Jeff Brozena**, Johnna Blair, Thomas Richardson, Mark Matthews, Dahlia Mukherjee, Erika F. H. Saunders, Saeed Abdullah. 2024. Supportive Fintech for Individuals with Bipolar Disorder: Financial Data Sharing Preferences to Support Longitudinal Care Management. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24), May 11–16, 2024, Honolulu, HI, USA*. ACM, New York, NY, USA. https://doi.org/10.1145/3613904.3642645.

Journal Articles

2025 Emma Morton, Elsy Willis, **Jeff Brozena**, Andrew Kcomt, Erin E. Michalak. 2025. The Type, Impacts, and Experiences of Peer Support for People Living With Bipolar Disorder: A Scoping Review. *Bipolar Disorders*. https://doi.org/10.1111/bdi.70006.

2022 Johnna Blair, **Jeff Brozena**, Mark Matthews, Thomas Richardson, Saeed Abdullah. Financial technologies (FinTech) for mental health: The potential of objective financial data to better understand the relationships between financial behavior and mental health. *Front. Psychiatry* 13:810057. https://doi.org/10.3389/fpsyt.2022.810057.

Presentations

2026 **Jeff Brozena**, Johnna Blair, Dahlia Mukherjee, Erika F.H. Saunders, M.D., Saeed Abdullah. 2026. Mapping Financial Behavior in Bipolar Disorder: Development of a Dataset Linking Spending Habits to Mood Fluctuations. Poster presented at the 28th Annual Conference of the International Society for Bipolar Disorders, Vancouver, Canada, September 2026; Poster presented at Penn State College of Medicine Psychiatry and Behavioral Health Research Conference, Hershey, PA, April 2026. https://doi.org/10.5281/zenodo.23114162. **Awarded Exemplary Graduate Student Research Poster**.

2026 **Jeff Brozena**, Saeed Abdullah. Evidence-based digital design: Utilizing MaxDiff findings to guide the development of financial interventions in bipolar disorder. Poster presented at the 28th Annual Conference of the International Society for Bipolar Disorders, Vancouver, Canada, September 2026. https://doi.org/10.5281/zenodo.23114255.

2026 **Jeff Brozena**, Saeed Abdullah, Lan Kong, Erika F.H. Saunders, Guodong Liu. Eight Years of Autonomic Monitoring: An N-of-1 Longitudinal Study of Wearable-derived HRV Anomalies and Self-reported Mood Logs. Poster presented at the 28th Annual Conference of the International Society for Bipolar Disorders, Vancouver, Canada, September 2026. https://doi.org/10.5281/zenodo.23193986.

Workshops

2026 Chris Elsden, Belen Barros Pena, Helena Lyhme, **Jeff Brozena**, Daniel Mwesigwa, Chris Speed, Jofish Kaye, John Vines. Configuring Money as an Interface. In Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI EA '26). Association for Computing Machinery, New York, NY, USA. https://moneyasinterface.designinformatics.org/workshop_paper.pdf.

2025 Johnna Blair, **Jeff Brozena**, John Vines, Jofish Kaye, Mark Matthews, Saeed Abdullah. 2025. Future of Money and HCI. In Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25). Association for Computing Machinery, New York, NY, USA, Article 780, 1–6. https://doi.org/10.1145/3706599.3706711.

Professional Affiliations

2023 Association for Computing Machinery

2023 CrestBD Network Member

2022 Society for Digital Mental Health

Teaching

2022 Teaching assistant, Methods for Studying Users (HCDD 364w)

2022 Teaching assistant, Mobile Application Design and Development (HCDD 340)

2021 Teaching assistant, Application Development Design Studio (IST 261)